KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5493C | 5 Credits: 4 | Program Elective | Source-grounded

Programming with VHDL

 To explore VHDL and its capabilities To learn to use VHDL design units

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5493C is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5493C.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Microelectronics
Course code	5493C
Course title	Programming with VHDL
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

14 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To explore VHDL and its capabilities To learn to use VHDL design units
- To build VHDL models using different modeling styles

Course outcomes

CO	Official outcome	Study level
CO1	Understand VHDL design units and language 15 Understanding elements Develop digital circuits by applying dataflow	See official syllabus
CO2	14 Applying modeling Apply structural modeling to develop digital	See official syllabus
CO3	14 Applying electronics circuits	See official syllabus
CO4	Develop digital designs using behavioral modeling 15 Applying	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 3 3
CO2	CO2 3 3 3
CO3	CO3 3 3 3
CO4	CO4 3 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Understand VHDL design units and language elements	4	As specified
Module 2	Develop digital circuits by applying dataflow modeling	4	As specified
Module 3	Apply structural modeling to develop digital electronics circuits	3	As specified
Module 4	Develop digital designs using behavioral modeling	3	As specified

MODULE 1

Understand VHDL design units and language elements

This module is presented from the official syllabus scope for Programming with VHDL. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand VHDL design units and language elements

- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the capabilities of VHDL

Explain the capabilities of VHDL is an official topic in Module 1 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the capabilities of VHDL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the capabilities of vhdl should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the capabilities of VHDL means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain the capabilities of VHDL [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Discuss different design units in VHDL

Discuss different design units in VHDL is an official topic in Module 1 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss different design units in VHDL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss different design units in vhdl should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss different design units in VHDL means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Discuss different design units in VHDL [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain different data types in VHDL

Explain different data types in VHDL is an official topic in Module 1 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different data types in VHDL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different data types in vhdl should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different data types in VHDL means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain different data types in VHDL [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Discuss the application of operators in VHDL 3 Understanding Introduction VHDL - History, Capabilities, Hardware Abstraction, Basic Terminology - Entity Declaration, Architecture Body - Structural Style of Modeling, Dataflow Style of Modeling, Behavioral Style of Modeling, Configuration Declaration, Package Declaration, Package Body, Model Analysis, Simulation. Basic Language Elements Identifiers, Data Objects, Data Types - Subtypes, Scalar Types, Composite Types, Access Types, Incomplete Types, File Types, Operators - Logical Operators, Relational Operators, Adding Operators, Multiplying Operators, Miscellaneous Operators

Discuss the application of operators in VHDL 3 Understanding Introduction VHDL - History, Capabilities, Hardware Abstraction, Basic Terminology - Entity Declaration, Architecture Body - Structural Style of Modeling, Dataflow Style of Modeling, Behavioral Style of Modeling, Configuration Declaration, Package Declaration, Package Body, Model Analysis, Simulation. Basic Language Elements Identifiers, Data Objects, Data Types - Subtypes, Scalar Types, Composite Types, Access Types, Incomplete Types, File Types, Operators - Logical Operators, Relational Operators, Adding Operators, Multiplying Operators, Miscellaneous Operators is an official topic in Module 1 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the application of operators in VHDL 3 Understanding Introduction VHDL - History, Capabilities, Hardware Abstraction, Basic Terminology - Entity Declaration, Architecture Body - Structural Style of Modeling, Dataflow Style of Modeling, Behavioral Style of Modeling, Configuration Declaration, Package Declaration, Package Body, Model Analysis, Simulation. Basic Language Elements Identifiers, Data Objects, Data Types - Subtypes, Scalar Types, Composite Types, Access Types, Incomplete Types, File Types, Operators - Logical Operators, Relational Operators, Adding Operators, Multiplying Operators, Miscellaneous Operators using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the application of operators in vhdl 3 understanding introduction vhdl - history, capabilities, hardware abstraction, basic terminology - entity declaration, architecture body - structural style of modeling, dataflow style of modeling, behavioral style of modeling, configuration declaration, package declaration, package body, model analysis, simulation. basic language elements identifiers, data objects, data types - subtypes, scalar types, composite types, access types, incomplete types, file types, operators - logical operators, relational operators, adding operators, multiplying operators, miscellaneous operators should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the application of operators in VHDL 3 Understanding Introduction VHDL - History, Capabilities, Hardware Abstraction, Basic Terminology - Entity Declaration, Architecture Body - Structural Style of Modeling, Dataflow Style of Modeling, Behavioral Style of Modeling, Configuration Declaration, Package Declaration, Package Body, Model Analysis, Simulation. Basic Language Elements Identifiers, Data Objects, Data Types - Subtypes, Scalar Types, Composite Types, Access Types, Incomplete Types, File Types, Operators - Logical Operators, Relational Operators, Adding Operators, Multiplying Operators, Miscellaneous Operators means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Discuss the application of operators in VHDL 3 Understanding Introduction VHDL - History, Capabilities, Hardware Abstraction, Basic Terminology - Entity Declaration, Architecture Body - Structural Style of Modeling, Dataflow Style of Modeling, Behavioral Style of Modeling, Configuration Declaration, Package Declaration, Package Body, Model Analysis, Simulation. Basic Language Elements Identifiers, Data Objects, Data Types - Subtypes, Scalar Types, Composite Types, Access Types, Incomplete Types, File Types, Operators - Logical Operators, Relational Operators, Adding Operators, Multiplying Operators, Miscellaneous Operators [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Understand VHDL design units and language elements

  M1.01      Explain the capabilities of VHDL                   2
Understanding

  M1.02      Discuss different design units in VHDL             5
Understanding

  M1.03      Explain different data types in VHDL               5
Understanding

  M1.04      Discuss the application of operators in VHDL       3
Understanding

Contents:
Introduction
VHDL - History, Capabilities, Hardware Abstraction, Basic Terminology - Entity
Declaration, Architecture Body - Structural Style of Modeling, Dataflow Style of
Modeling, Behavioral Style of Modeling, Configuration Declaration, Package
Declaration,
Package Body, Model Analysis, Simulation.
Basic Language Elements
Identifiers, Data Objects, Data Types - Subtypes, Scalar Types, Composite Types,
Access
Types, Incomplete Types, File Types, Operators - Logical Operators, Relational
Operators,
Adding Operators, Multiplying Operators, Miscellaneous Operators
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Develop digital circuits by applying dataflow modeling

This module is presented from the official syllabus scope for Programming with VHDL. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Develop digital circuits by applying dataflow modeling
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Applying signal assignment Develop a digital design using conditional

3 Applying signal assignment Develop a digital design using conditional is an official topic in Module 2 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying signal assignment Develop a digital design using conditional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying signal assignment develop a digital design using conditional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying signal assignment Develop a digital design using conditional means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-[illegible] 3 Applying signal assignment Develop a digital design using conditional [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Applying and selected signal assignment statement Apply block statement and assertion

3 Applying and selected signal assignment statement Apply block statement and assertion is an official topic in Module 2 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying and selected signal assignment statement Apply block statement and assertion using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying and selected signal assignment statement apply block statement and assertion should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying and selected signal assignment statement Apply block statement and assertion means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 3 Applying and selected signal assignment statement Apply block statement and assertion [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Applying statement in digital logic design Develop digital designs using dataflow

4 Applying statement in digital logic design Develop digital designs using dataflow is an official topic in Module 2 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying statement in digital logic design Develop digital designs using dataflow using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying statement in digital logic design develop digital designs using dataflow should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying statement in digital logic design Develop digital designs using dataflow means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Applying statement in digital logic design Develop digital designs using dataflow □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Applying modeling Dataflow Modeling Concurrent Signal Assignment Statement, Concurrent versus Sequential Signal Assignment, Delta Delay Revisited, Multiple Drivers, Conditional Signal Assignment Statement, Selected Signal Assignment Statement, Block Statement, Concurrent Assertion Statement Programming Examples : Logic gates, Half adder, Full adder, 4:1 Multiplexer, 2-bit Comparator

4 Applying modeling Dataflow Modeling Concurrent Signal Assignment Statement, Concurrent versus Sequential Signal Assignment, Delta Delay Revisited, Multiple Drivers, Conditional Signal Assignment Statement, Selected Signal Assignment Statement, Block Statement, Concurrent Assertion Statement Programming Examples : Logic gates, Half adder, Full adder, 4:1 Multiplexer, 2-bit Comparator is an official topic in Module 2 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying modeling Dataflow Modeling Concurrent Signal Assignment Statement, Concurrent versus Sequential Signal Assignment, Delta Delay Revisited, Multiple Drivers, Conditional Signal Assignment Statement, Selected Signal Assignment Statement, Block Statement, Concurrent Assertion Statement Programming Examples : Logic gates, Half adder, Full adder, 4:1 Multiplexer, 2-bit Comparator using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying modeling dataflow modeling concurrent signal assignment statement, concurrent versus sequential signal assignment, delta delay revisited, multiple drivers, conditional signal assignment statement, selected signal assignment statement, block statement, concurrent assertion statement programming examples : logic gates, half adder, full adder, 4:1 multiplexer, 2-bit comparator should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying modeling Dataflow Modeling Concurrent Signal Assignment Statement, Concurrent versus Sequential Signal Assignment, Delta Delay Revisited, Multiple Drivers, Conditional Signal Assignment Statement, Selected Signal Assignment Statement, Block Statement, Concurrent Assertion Statement Programming Examples : Logic gates, Half adder, Full adder, 4:1 Multiplexer, 2-bit Comparator means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Applying modeling Dataflow Modeling Concurrent Signal Assignment Statement, Concurrent versus Sequential Signal Assignment, Delta Delay Revisited, Multiple Drivers, Conditional Signal Assignment Statement, Selected Signal Assignment Statement, Block Statement, Concurrent Assertion Statement Programming Examples : Logic gates, Half adder, Full adder, 4:1 Multiplexer, 2-bit Comparator □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Develop digital circuits by applying dataflow modeling

             Implement digital logic using concurrent
  M2.01                                                         3
Applying
             signal assignment
             Develop a digital design using conditional
  M2.02                                                         3
Applying
             and selected signal assignment statement
             Apply block statement and assertion
  M2.03                                                         4
Applying
             statement in digital logic design
            Develop digital    designs   using   dataflow
  M2.04                                                        4
Applying
            modeling
            Series Test - I                                    1

Contents:
Dataflow Modeling
Concurrent Signal Assignment Statement, Concurrent versus Sequential Signal
Assignment, Delta Delay Revisited, Multiple Drivers, Conditional Signal
Assignment
Statement, Selected Signal Assignment Statement, Block Statement, Concurrent
Assertion
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Apply structural modeling to develop digital electronics circuits

This module is presented from the official syllabus scope for Programming with VHDL. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Apply structural modeling to develop digital electronics circuits
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

4 Applying and component instantiation

4 Applying and component instantiation is an official topic in Module 3 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying and component instantiation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying and component instantiation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying and component instantiation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 4 Applying and component instantiation [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Understand resolution function 2 Understanding Design digital electronics circuits using

Understand resolution function 2 Understanding Design digital electronics circuits using is an official topic in Module 3 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand resolution function 2 Understanding Design digital electronics circuits using using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand resolution function 2 understanding design digital electronics circuits using should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand resolution function 2 Understanding Design digital electronics circuits using means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Understand resolution function 2 Understanding Design digital electronics circuits using □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

8 Applying structural modeling Structural Modeling Component Declaration, Component Instantiation, Resolving Signal Values Programming Examples : Half adder using gates, Full adder using half adder, 4-bit Ripple Carry Adder, 4:1 multiplexer using 2:1 mux, D-FF, JK-FF, T-FF

8 Applying structural modeling Structural Modeling Component Declaration, Component Instantiation, Resolving Signal Values Programming Examples : Half adder using gates, Full adder using half adder, 4-bit Ripple Carry Adder, 4:1 multiplexer using 2:1 mux, D-FF, JK-FF, T-FF is an official topic in Module 3 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Applying structural modeling Structural Modeling Component Declaration, Component Instantiation, Resolving Signal Values Programming Examples : Half adder using gates, Full adder using half adder, 4-bit Ripple Carry Adder, 4:1 multiplexer using 2:1 mux, D-FF, JK-FF, T-FF using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 applying structural modeling structural modeling component declaration, component instantiation, resolving signal values programming examples : half adder using gates, full adder using half adder, 4-bit ripple carry adder, 4:1 multiplexer using 2:1 mux, d-ff, jk-ff, t-ff should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 8 Applying structural modeling Structural Modeling Component Declaration, Component Instantiation, Resolving Signal Values Programming Examples : Half adder using gates, Full adder using half adder, 4-bit Ripple Carry Adder, 4:1 multiplexer using 2:1 mux, D-FF, JK-FF, T-FF means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 8 Applying structural modeling Structural Modeling Component Declaration, Component Instantiation, Resolving Signal Values Programming Examples : Half adder using gates, Full adder using half adder, 4-bit Ripple Carry Adder, 4:1 multiplexer using 2:1 mux, D-FF, JK-FF, T-FF □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Apply structural modeling to develop digital electronics circuits

            Model a design using component declaration
  M3.01                                                        4           Applying
            and component instantiation
  M3.02     Understand resolution function                     2
Understanding
            Design digital electronics circuits using
  M3.03                                                        8           Applying
            structural modeling
Contents:
Structural Modeling
Component Declaration, Component Instantiation, Resolving Signal Values
Programming Examples : Half adder using gates, Full adder using half adder, 4-bit
Ripple
Carry Adder, 4:1 multiplexer using 2:1 mux, D-FF, JK-FF, T-FF
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Develop digital designs using behavioral modeling

This module is presented from the official syllabus scope for Programming with VHDL. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Develop digital designs using behavioral modeling
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

2 Applying entity using process Implement various sequential statements in

2 Applying entity using process Implement various sequential statements in is an official topic in Module 4 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying entity using process Implement various sequential statements in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying entity using process implement various sequential statements in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying entity using process Implement various sequential statements in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-[illegible] 2 Applying entity using process Implement various sequential statements in [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

6 Applying behavioral modeling Design digital circuits using behavioral

6 Applying behavioral modeling Design digital circuits using behavioral is an official topic in Module 4 of Programming with VHDL. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Applying behavioral modeling Design digital circuits using behavioral using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 applying behavioral modeling design digital circuits using behavioral should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Applying behavioral modeling Design digital circuits using behavioral means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ 6 Applying behavioral modeling Design digital circuits using behavioral □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

7 Applying modeling Behavioral Modeling Process Statement, Variable Assignment Statement, Signal Assignment Statement, Wait Statement, If Statement, Case Statement, Null Statement, Loop Statement, Exit Statement, Next Statement, Assertion Statement, Other Sequential Statements, Multiple Processes. Programming Examples : 8:1 Multiplexer, D-FF, JK FF, T FF, 3-bit upcounter, 3-bit downcounter Text /Reference: T/R Book Title/Author T1 VHDL Primer Third edition by J. Bhaskar, PHI Brown S. and Vranesic Z., Fundamentals of Digital Logic with VHDL Design, R1 2/e,TMH.2008 R2 Perry D. L., VHDL Programming by Example, 4/e, TMH, 2008. R3 Roth C.H., Digital System Design Using VHDL, Cengage Learning, 2008 R4 Pedroni V. A., Circuit design with VHDL, PHI, 2008. R5 Kevin Skahill, VHDL for Programmable Logic, Addison & Wesley, 1996 Online Resources: Sl. No. Website Link https://www.edutechlearners.com/download/books/A%20VHDL%20Primer%20 1 -%20Jayaram%20Bhasker.pdf https://www.intel.com/content/www/us/en/programmable/customertraining/webe 2 x/VHDL/presentation_html5.html https://books.google.co.in/books?id=KQvx_b0C9BIC&printsec=frontcover#v=o 3 nepage&q&f=false 4 https://www.youtube.com/@synthesisofdigitalsystems-5267/videos 5 https://www.udemy.com/course/vhdl-programming-for-beginners/

7 Applying modeling Behavioral Modeling Process Statement, Variable Assignment Statement, Signal Assignment Statement, Wait Statement, If Statement, Case Statement, Null Statement, Loop Statement, Exit Statement, Next Statement, Assertion Statement, Other Sequential Statements, Multiple Processes. Programming Examples : 8:1 Multiplexer, D-FF, JK FF, T FF, 3-bit upcounter, 3-bit downcounter Text /Reference: T/R Book Title/Author T1 VHDL Primer Third edition by J. Bhaskar, PHI Brown S. and Vranesic Z., Fundamentals of Digital Logic with VHDL Design, R1 2/e,TMH.2008 R2 Perry D. L., VHDL Programming by Example, 4/e, TMH, 2008. R3 Roth C.H., Digital System Design Using VHDL, Cengage Learning, 2008 R4 Pedroni V. A., Circuit design with VHDL, PHI, 2008. R5 Kevin Skahill, VHDL for Programmable Logic, Addison & Wesley, 1996 Online Resources: Sl. No. Website Link https://www.edutechlearners.com/download/books/A%20VHDL%20Primer%20 1 -%20Jayaram%20Bhasker.pdf https://www.intel.com/content/www/us/en/programmable/customertraining/webe 2 x/VHDL/presentation_html5.html https://books.google.co.in/books?id=KQvx_b0C9BIC&printsec=frontcover#v=o 3 nepage&q&f=false 4 https://www.youtube.com/@synthesisofdigitalsystems-5267/videos 5 https://www.udemy.com/course/vhdl-programming-for-beginners/ is an official topic in Module 4 of Programming with VHDL. Study the terminology first, then

connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Applying modeling Behavioral Modeling Process Statement, Variable Assignment Statement, Signal Assignment Statement, Wait Statement, If Statement, Case Statement, Null Statement, Loop Statement, Exit Statement, Next Statement, Assertion Statement, Other Sequential Statements, Multiple Processes. Programming Examples : 8:1 Multiplexer, D-FF, JK FF, T FF, 3-bit upcounter, 3-bit downcounter Text /Reference: T/R Book Title/Author T1 VHDL Primer Third edition by J. Bhaskar, PHI Brown S. and Vranesic Z., Fundamentals of Digital Logic with VHDL Design, R1 2/e,TMH.2008 R2 Perry D. L., VHDL Programming by Example, 4/e, TMH, 2008. R3 Roth C.H., Digital System Design Using VHDL, Cengage Learning, 2008 R4 Pedroni V. A., Circuit design with VHDL, PHI, 2008. R5 Kevin Skahill, VHDL for Programmable Logic, Addison & Wesley, 1996 Online Resources: Sl. No. Website Link https://www.edutechlearners.com/download/books/A%20VHDL%20Primer%20 1 -%20Jayaram%20Bhasker.pdf https://www.intel.com/content/www/us/en/programmable/customertraining/webe 2 x/VHDL/presentation_html5.html https://books.google.co.in/books?id=KQvx_b0C9BIC&printsec=frontcover#v=o 3 nepage&q&f=false 4 https://www.youtube.com/@synthesisofdigitalsystems-5267/videos 5 https://www.udemy.com/course/vhdl-programming-for-beginners/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 applying modeling behavioral modeling process statement, variable assignment statement, signal assignment statement, wait statement, if statement, case statement, null statement, loop statement, exit statement, next statement, assertion statement, other sequential statements, multiple processes. programming examples : 8:1 multiplexer, d-ff, jk ff, t ff, 3-bit upcounter, 3-bit downcounter text /reference: t/r book title/author t1 vhdl primer third edition by j.

bhaskar, phi brown s. and vranesic z., fundamentals of digital logic with vhdl design, r1 2/e,tmh.2008 r2 perry d. l., vhdl programming by example, 4/e, tmh, 2008. r3 roth c.h., digital system design using vhdl, cengage learning, 2008 r4 pedroni v. a., circuit design with vhdl, phi, 2008. r5 kevin skahill, vhdl for programmable logic, addison & wesley, 1996 online resources: sl. no. website link https://www.edutechlearners.com/download/books/a%20vhdl%20primer%20 1 - %20jayaram%20bhasker.pdf https://www.intel.com/content/www/us/en/programmable/customertraining/webe 2 x/vhdl/presentation_html5.html https://books.google.co.in/books?id=kqvx_b0c9bic&printsec=frontcover#v=o 3 nepage&q&f=false 4 https://www.youtube.com/@synthesisofdigitalsystems-5267/videos 5 https://www.udemy.com/course/vhdl-programming-for-beginners/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Applying modeling Behavioral Modeling Process Statement, Variable Assignment Statement, Signal Assignment Statement, Wait Statement, If Statement, Case Statement, Null Statement, Loop Statement, Exit Statement, Next Statement, Assertion Statement, Other Sequential Statements, Multiple Processes. Programming Examples : 8:1 Multiplexer, D-FF, JK FF, T FF, 3-bit upcounter, 3-bit downcounter Text /Reference: T/R Book Title/Author T1 VHDL Primer Third edition by J. Bhaskar, PHI Brown S. and Vranesic Z., Fundamentals of Digital Logic with VHDL Design, R1 2/e,TMH.2008 R2 Perry D. L., VHDL Programming by Example, 4/e, TMH, 2008. R3 Roth C.H., Digital System Design Using VHDL, Cengage Learning, 2008 R4 Pedroni V. A., Circuit design with VHDL, PHI, 2008. R5 Kevin Skahill, VHDL for Programmable Logic, Addison & Wesley, 1996 Online Resources: Sl. No. Website Link https://www.edutechlearners.com/download/books/A%20VHDL%20Primer%20 1 - %20Jayaram%20Bhasker.pdf https://www.intel.com/content/www/us/en/programmable/customertraining/webe 2 x/VHDL/presentation_html5.html https://books.google.co.in/books?id=KQvx_b0C9BIC&printsec=frontcover#v=o 3 nepage&q&f=false 4 https://www.youtube.com/@synthesisofdigitalsystems-5267/videos 5 https://www.udemy.com/course/vhdl-programming-for-beginners/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 7 Applying modeling Behavioral Modeling Process Statement, Variable Assignment Statement, Signal Assignment Statement, Wait Statement, If Statement, Case Statement, Null Statement, Loop Statement, Exit Statement, Next Statement, Assertion Statement, Other Sequential Statements, Multiple Processes. Programming Examples : 8:1 Multiplexer, D-FF, JK FF, T FF, 3-bit upcounter, 3-bit downcounter Text /Reference: T/R Book Title/Author T1 VHDL Primer Third edition by J. Bhaskar, PHI Brown S. and Vranesic Z., Fundamentals of Digital Logic with VHDL Design, R1 2/e,TMH.2008 R2 Perry D. L., VHDL Programming by Example, 4/e, TMH, 2008. R3 Roth C.H., Digital System Design Using VHDL, Cengage Learning, 2008 R4 Pedroni V. A., Circuit design with VHDL, PHI, 2008. R5 Kevin Skahill, VHDL for Programmable Logic, Addison & Wesley, 1996 Online Resources: Sl. No. Website Link https://www.edutechlearners.com/download/books/A%20VHDL%20Primer%20 1 -%20Jayaram%20Bhasker.pdf https://www.intel.com/content/www/us/en/programmable/customertraining/webe 2 x/VHDL/presentation_html5.html https://books.google.co.in/books?id=KQvx_b0C9BIC&printsec=frontcover#v=o 3 nepage&q&f=false 4 https://www.youtube.com/@synthesisofdigitalsystems-5267/videos 5 https://www.udemy.com/course/vhdl-programming-for-beginners/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Develop digital designs using behavioral modeling

            Model the procedural type behavior of an
  M4.01                                                        2
Applying
            entity using process
            Implement various sequential statements in
  M4.02                                                        6
Applying
            behavioral modeling

            Design digital circuits using behavioral
  M4.03                                                        7
Applying
            modeling

            Series Test - I                                    1

Contents:

Behavioral Modeling
Process Statement, Variable Assignment Statement, Signal Assignment Statement,
Wait
Statement, If Statement, Case Statement, Null Statement, Loop Statement, Exit
Statement,
Next Statement, Assertion Statement, Other Sequential Statements, Multiple
Processes.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain the capabilities of VHDL. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the capabilities of VHDL*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Discuss different design units in VHDL. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss different design units in VHDL*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain different data types in VHDL. (3 marks)

Answer: Begin with the standard definition or identity of *Explain different data types in VHDL*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Discuss the application of operators in VHDL 3 Understanding Introduction VHDL - History, Capabilities, Hardware Abstraction, Basic Terminology - Entity Declaration, Architecture Body - Structural Style of Modeling, Dataflow Style of Modeling, Behavioral Style of Modeling, Configuration Declaration, Package Declaration, Package Body, Model Analysis, Simulation. Basic Language Elements Identifiers, Data Objects, Data Types - Subtypes, Scalar Types, Composite Types, Access Types, Incomplete Types, File Types, Operators - Logical Operators, Relational Operators, Adding Operators, Multiplying Operators, Miscellaneous Operators. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss the application of operators in VHDL 3 Understanding Introduction VHDL - History, Capabilities, Hardware Abstraction, Basic Terminology - Entity Declaration, Architecture Body - Structural Style* *of Modeling, Dataflow Style of Modeling, Behavioral Style of Modeling, Configuration Declaration, Package Declaration, Package Body, Model Analysis, Simulation. Basic Language Elements Identifiers, Data Objects, Data Types - Subtypes, Scalar Types, Composite Types, Access Types, Incomplete Types, File Types, Operators - Logical Operators, Relational Operators, Adding Operators, Multiplying Operators, Miscellaneous Operators*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 3 Applying signal assignment Develop a digital design using conditional. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying signal assignment Develop a digital design using conditional*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Applying and selected signal assignment statement Apply block statement and assertion. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying and selected signal assignment statement Apply block statement and assertion*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Applying statement in digital logic design Develop digital

designs using dataflow. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying statement in digital logic design Develop digital designs using dataflow*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Applying modeling Dataflow Modeling Concurrent Signal Assignment Statement, Concurrent versus Sequential Signal Assignment, Delta Delay Revisited, Multiple Drivers, Conditional Signal Assignment Statement, Selected Signal Assignment Statement, Block Statement, Concurrent Assertion Statement Programming Examples : Logic gates, Half adder, Full adder, 4:1 Multiplexer, 2-bit Comparator. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying modeling Dataflow Modeling Concurrent Signal Assignment Statement, Concurrent versus Sequential Signal Assignment, Delta Delay Revisited, Multiple Drivers, Conditional Signal Assignment Statement, Selected Signal Assignment Statement, Block Statement, Concurrent Assertion Statement Programming Examples : Logic gates, Half adder, Full adder, 4:1 Multiplexer, 2-bit Comparator*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 4 Applying and component instantiation. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying and component instantiation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Understand resolution function 2 Understanding Design digital electronics circuits using. (3 marks)

Answer: Begin with the standard definition or identity of *Understand resolution function 2 Understanding Design digital electronics circuits using*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 8 Applying structural modeling Structural Modeling Component Declaration, Component Instantiation, Resolving Signal Values Programming Examples : Half adder using gates, Full adder using half adder, 4-bit Ripple Carry Adder, 4:1 multiplexer using 2:1 mux, D-FF, JK-FF, T-FF. (3 marks)

Answer: Begin with the standard definition or identity of *8 Applying structural modeling Structural Modeling Component Declaration, Component Instantiation, Resolving Signal Values Programming Examples : Half adder using gates, Full adder using half adder, 4-bit Ripple Carry Adder, 4:1 multiplexer using 2:1 mux, D-FF, JK-FF, T-FF*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 2 Applying entity using process Implement various sequential statements in. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying entity using process Implement various sequential statements in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 6 Applying behavioral modeling Design digital circuits using behavioral. (3 marks)

Answer: Begin with the standard definition or identity of *6 Applying behavioral modeling Design digital circuits using behavioral*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 7 Applying modeling Behavioral Modeling Process Statement, Variable Assignment Statement, Signal Assignment Statement, Wait Statement, If Statement, Case Statement, Null Statement, Loop Statement, Exit Statement, Next Statement, Assertion Statement, Other Sequential Statements, Multiple Processes. Programming Examples : 8:1 Multiplexer, D-FF, JK FF, T FF, 3-bit upcounter, 3-bit downcounter Text /Reference: T/R Book Title/Author T1 VHDL Primer Third edition by J. Bhaskar, PHI Brown S. and Vranesic Z., Fundamentals of Digital Logic with VHDL Design, R1 2/e,TMH.2008 R2 Perry D. L., VHDL Programming by Example, 4/e, TMH, 2008. R3 Roth C.H., Digital System Design Using VHDL, Cengage Learning, 2008 R4 Pedroni V. A., Circuit design with VHDL, PHI, 2008. R5 Kevin Skahill, VHDL for Programmable Logic, Addison & Wesley, 1996 Online Resources: Sl. No. Website Link https://www.edutechlearners.com/download/books/A%20VHDL%20Primer%20 1 -%20Jayaram%20Bhasker.pdf https://www.intel.com/content/www/us/en/programmable/customertraining/webe 2 x/VHDL/presentation_html5.html https://books.google.co.in/books? id=KQvx_b0C9BIC&printsec=frontcover#v=o 3 nepage&q&f=false 4

https://www.youtube.com/@synthesisofdigitalsystems-5267/videos 5 https://www.udemy.com/course/vhdl-programming-for-beginners/. (3 marks)

Answer: Begin with the standard definition or identity of *7 Applying modeling Behavioral Modeling Process Statement, Variable Assignment Statement, Signal Assignment Statement, Wait Statement, If Statement, Case Statement, Null Statement, Loop Statement, Exit Statement, Next Statement, Assertion Statement, Other Sequential Statements, Multiple Processes. Programming Examples : 8:1 Multiplexer, D-FF, JK FF, T FF, 3-bit upcounter, 3-bit downcounter Text /Reference: T/R Book Title/Author T1 VHDL Primer Third edition by J. Bhaskar, PHI Brown S. and Vranesic Z., Fundamentals of Digital Logic with VHDL Design, R1 2/e,TMH.2008 R2 Perry D. L., VHDL Programming by Example, 4/e, TMH, 2008. R3 Roth C.H., Digital System Design Using VHDL, Cengage Learning, 2008 R4 Pedroni V. A., Circuit design with VHDL, PHI, 2008. R5 Kevin Skahill, VHDL for Programmable Logic, Addison & Wesley, 1996 Online Resources: Sl. No. Website Link https://www.edutechlearners.com/download/books/A%20VHDL%20Primer%20 1 -%20Jayaram%20Bhasker.pdf https://www.intel.com/content/www/us/en/programmable/customertraining/webe 2 x/VHDL/presentation_html5.html https://books.google.co.in/books?id=KQvx_b0C9BIC&printsec=frontcover#v=o 3 nepage&q&f=false 4 https://www.youtube.com/@synthesisofdigitalsystems-5267/videos 5 https://www.udemy.com/course/vhdl-programming-for-beginners/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** [illegible] definition, [illegible] principle/steps, [illegible] application [illegible].

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain the capabilities of VHDL**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Applying signal assignment Develop a digital design using conditional**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **4 Applying and component instantiation**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **2 Applying entity using process Implement various sequential statements in**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No. Website Link
- https://www.edutechlearners.com/download/books/A%20VHDL%20Primer%20%20Jayaram%20Bhasker.pdf
- https://www.intel.com/content/www/us/en/programmable/customertraining/webex/VHDL/presentation_html5.html
- https://books.google.co.in/books?id=KQvx_b0C9BIC&printsec=frontcover#v=onepage&q&f=false
- https://www.youtube.com/@synthesisofdigitalsystems-5267/videos
- https://www.udemy.com/course/vhdl-programming-for-beginners/

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